

PROJECT REPORT OF CO-OP PROJECT AT INDUSTRY (MODULE-1)

ON

SPOT

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

We hereby declare that the project work titled, SPOT submitted as part of Bachelor's degree in CSE, at Chitkara University, Punjab, is an authentic record of our own work carried out under the supervision of Dr. Swati Goel.

Signature(s):
Rishika Pawar

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Dr. Swati Goel**, my project supervisor, for her invaluable guidance, constant encouragement, and insightful feedback throughout the completion of my Integrated Project. Her dedication, expertise, and thoughtful suggestions were instrumental in shaping the direction, depth, and overall quality of this work. Her willingness to devote time and effort at every stage of the project greatly enhanced my learning experience and contributed significantly to my academic development.

I am also thankful to **Chitkara University** for providing a supportive academic environment and the opportunity to undertake this Integrated Project as an integral part of the curriculum. The resources and guidance provided by the institution played a vital role in the successful completion of this project.

Finally, I extend my heartfelt gratitude to my family and friends for their unwavering support, encouragement, and constructive feedback throughout the course of this project. Their motivation and belief in me were invaluable in bringing this work to fruition.

ABSTRACT

As part of my internship at Orange Business, I have been introduced to an enterprise-level project named SPOT within the Workplace/IT Operations domain. The SPOT platform focuses on remote software and device management, enabling IT teams to efficiently manage, monitor, and troubleshoot organizational devices without requiring physical access. Through this project, I am gaining practical exposure to remote software deployment, remote task execution, and system control used in real-world enterprise environments.

During the course of this internship, I am working with modern .NET-based web technologies, infrastructure management tools, and centralized management systems to understand how large organizations reduce downtime and improve operational efficiency. Currently, I am undergoing Knowledge Transfer (KT) sessions and familiarizing myself with the tools, technologies, and workflows involved in the SPOT project. This report has been prepared as an urgent academic requirement and provides an overview of the scope, methodology, tools, and expected outcomes of the SPOT project.

1. Introduction

1.1 Background

With the rapid growth of digital infrastructure in large organizations, managing a wide range of employee devices efficiently has become a major challenge for IT teams. Traditional methods of on-site troubleshooting and manual software installation are time-consuming, costly, and inefficient, especially in geographically distributed environments. To overcome these challenges, organizations rely on centralized remote management platforms that enable automation, monitoring, and remote support.

Orange Business, requires robust and secure systems to manage enterprise devices and ensure uninterrupted business operations. In this context, the **SPOT** project plays a crucial role by providing a centralized solution for remote software management and device control.

1.2 Project Overview

As part of my internship at **Orange Business**, I have been introduced to an enterprise-level project named **SPOT**, which is designed to support IT operations and workplace management teams. SPOT is an enterprise remote software and device management platform that enables IT teams to remotely deploy applications, execute maintenance tasks, and troubleshoot systems without requiring physical access to devices.

The platform supports a wide range of organizational devices such as office computers, servers, cash registers, and digital display systems. By using SPOT, IT teams can respond quickly to technical issues, reduce system downtime, and improve overall operational efficiency.

1.3 Purpose of the Project

The primary purpose of the SPOT project is to simplify and automate routine IT management tasks through a centralized web-based system. The platform allows authorized users to:

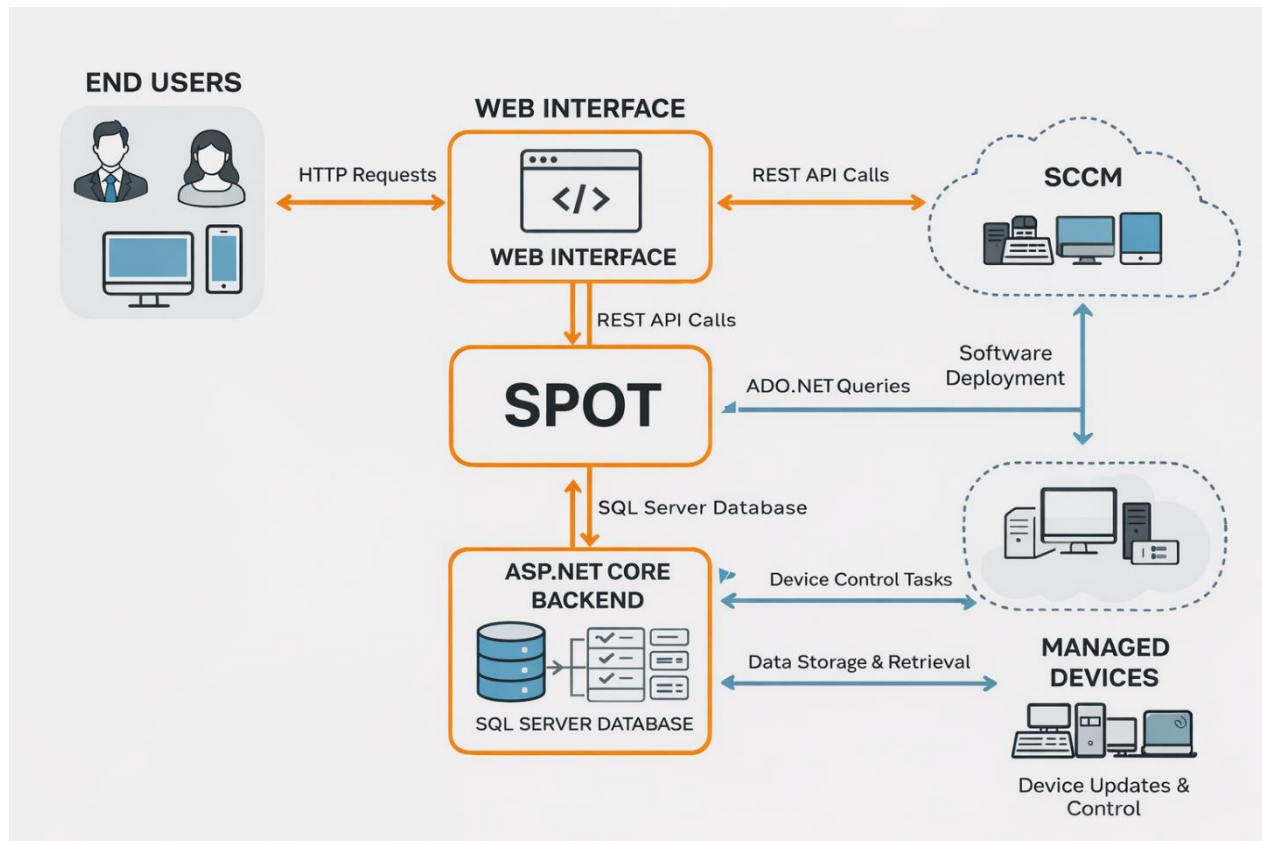
- Install and update software remotely on employee devices
- Execute troubleshooting and maintenance scripts
- Remotely control devices for issue diagnosis
- Manage different types of enterprise hardware from a single platform

This approach significantly reduces manual effort and improves response time for IT support teams.

1.4 Internship Learning Context

During the course of this internship, I am gaining hands-on exposure to real-world enterprise environments, modern **ASP.NET Core-based web application development**, and infrastructure management tools. I am also undergoing **Knowledge Transfer (KT) sessions**, which help in understanding the system architecture, workflows, and technologies used in the SPOT project.

This report has been prepared as an **urgent academic requirement** and documents the understanding gained regarding the project's objectives, methodology, design, implementation, and expected outcomes.



2. System Requirements

2.1 Hardware Requirements

- **Processor:** Intel i5 or higher
- **RAM:** Minimum 8 GB
- **Storage:** Minimum 256 GB SSD
- **Network:** Stable Internet connection
- **Devices:** Enterprise PCs, servers, and supported endpoints

2.2 Software Requirements

- **Operating System:** Windows 11
- **Backend Framework:** ASP.NET Core
- **Frontend Framework:** Angular
- **Database:** Microsoft SQL Server
- **Web Server:** IIS (Internet Information Services)

- **Deployment Tool: SCCM**
- **Development Tools: Visual Studio (2022), SSMS**
- **Version Control: GitLab**

3. Methodology

3.1 Agile Methodology

The SPOT project follows an Agile methodology, which focuses on iterative development and continuous improvement. Agile allows the team to adapt to changing requirements and deliver features incrementally.

3.2 Sprint Planning

The work is divided into sprints, each lasting around 2–3 weeks. Before each sprint, a sprint planning meeting is conducted where:

- Tasks are discussed and prioritized
- Sprint goals are defined
- Responsibilities are assigned to team members

3.3 Daily Scrum

Daily Scrum meetings are conducted to track progress. In these meetings, team members discuss:

- What was done the previous day
- What will be done today
- Any blockers or issues

3.4 Sprint Review and Retrospective

At the end of each sprint:

- A Sprint Review is conducted to demonstrate completed work
- A Sprint Retrospective is held to identify improvements for future sprints

4. SOFTWARE DESIGN

4.1 System Architecture

SPOT follows a **client-server architecture**. The frontend interacts with the backend services hosted on IIS. The backend communicates with SQL Server for data storage and SCCM for software deployment.

Major Components:

- Web Interface (Frontend)
- ASP.NET Core Backend
- SQL Server Database
- SCCM Integration
- IIS Web Server

4.2 Architecture Explanation

- Users access SPOT through a web interface
- Requests are processed by the ASP.NET Core backend
- Data is retrieved or stored using SQL Server via ADO.NET
- Software deployment tasks are handled using SCCM

4.3 UI/UX Design

The user interface is designed to be simple and user-friendly. It allows IT administrators to:

- View device details

- Deploy software
- Execute tasks remotely
- Monitor deployment status

For frontend, Angular is used.

5. IMPLEMENTATION DETAILS

5.1 Frameworks Used

- ASP.NET Core for backend development
- ADO.NET for database interaction
- IIS for application hosting

5.2 Database Design

SQL Server is used as the database. The database-first approach is followed using ADO.NET. Tables store information related to:

- Devices
- Users
- Software packages
- Deployment logs

5.3 Core Logic

The backend handles:

- Authentication and authorization
- Request validation

- Interaction with SCCM
- Data management operations

6. TESTING AND VALIDATION

Testing is performed to ensure system reliability and correctness.

6.1 Types of Testing

- Unit Testing: To test individual modules
- Integration Testing: To verify interaction between components
- Functional Testing: To ensure requirements are met

6.2 Validation

Test cases are executed to verify:

- Successful software deployment
- Proper task execution
- Secure access to the system

7. MAJOR FINDINGS / OUTCOMES

The project resulted in:

- Improved understanding of enterprise IT management
- Hands-on experience with ASP.NET Core and SQL Server
- Understanding of SCCM-based deployments
- Reduced manual effort for IT teams

- Faster issue resolution and improved efficiency

8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The SPOT project provides an effective solution for remote software and device management in enterprise environments. It enhances IT efficiency, reduces downtime, and simplifies device management.

8.2 Future Scope

- Cloud-based deployment
- Enhanced monitoring dashboards
- Automation using AI-driven analytics
- Support for additional device types

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APPENDICES

